Technische Hochschule Ingolstadt
Fakultät Elektro- und Informationstechnik

Final Presentation– Car2x Project

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- **Introduction to the Project**
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Introduction to V2X

V2X Technology

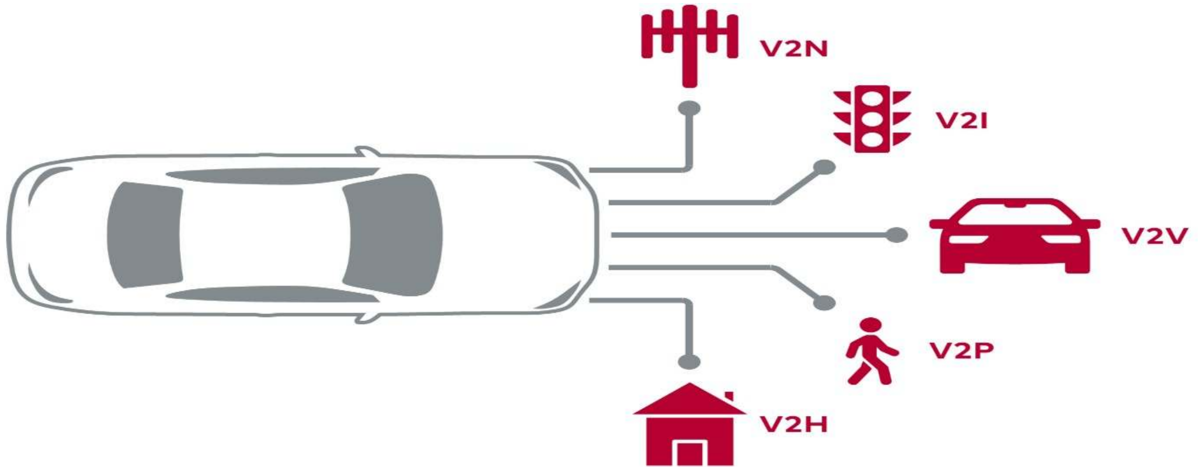
V2X (Vehicle-to-Everything) technology :

enables vehicles to communicate with:

- Vehicles
- Infrastructure
- Pedestrians
- users.

To enhance safety, efficiency, and mobility in transportation systems.

V2X utilizes wireless communication technologies to establish seamless connectivity and real-time data exchange.



V2X Communication Types:

Vehicle-to-Vehicle (V2V) Communication:

- direct communication between vehicles.
- exchange real-time information such as position, speed, and heading.

Vehicle-to-Infrastructure (V2I) Communication:

- communication between vehicles and roadside infrastructure elements. e.g. traffic lights, road signs

Vehicle-to-Pedestrian/Vehicle-to-User (V2P/V2U) Communication:

- communication between vehicles and pedestrians or other road users.

V2X Communication: Technical Overview

The two main technologies used in V2X are:

Dedicated Short-Range Communication (DSRC):

- DSRC operates in the 5.9 GHz frequency band based on IEEE 802.11 protocol

Cellular Vehicle-to-Everything (C-V2X):

- C-V2X utilizes existing cellular networks, such as 4G LTE and 5G, for V2X communication.
- It leverages the infrastructure and coverage of cellular networks for broader communication reach.

V2X Communication: Hardware Technologies

There are various hardware used for V2X

Cohda OBU unit

Commsignia RSU unit

Huawei RSU unit

Harman OBU

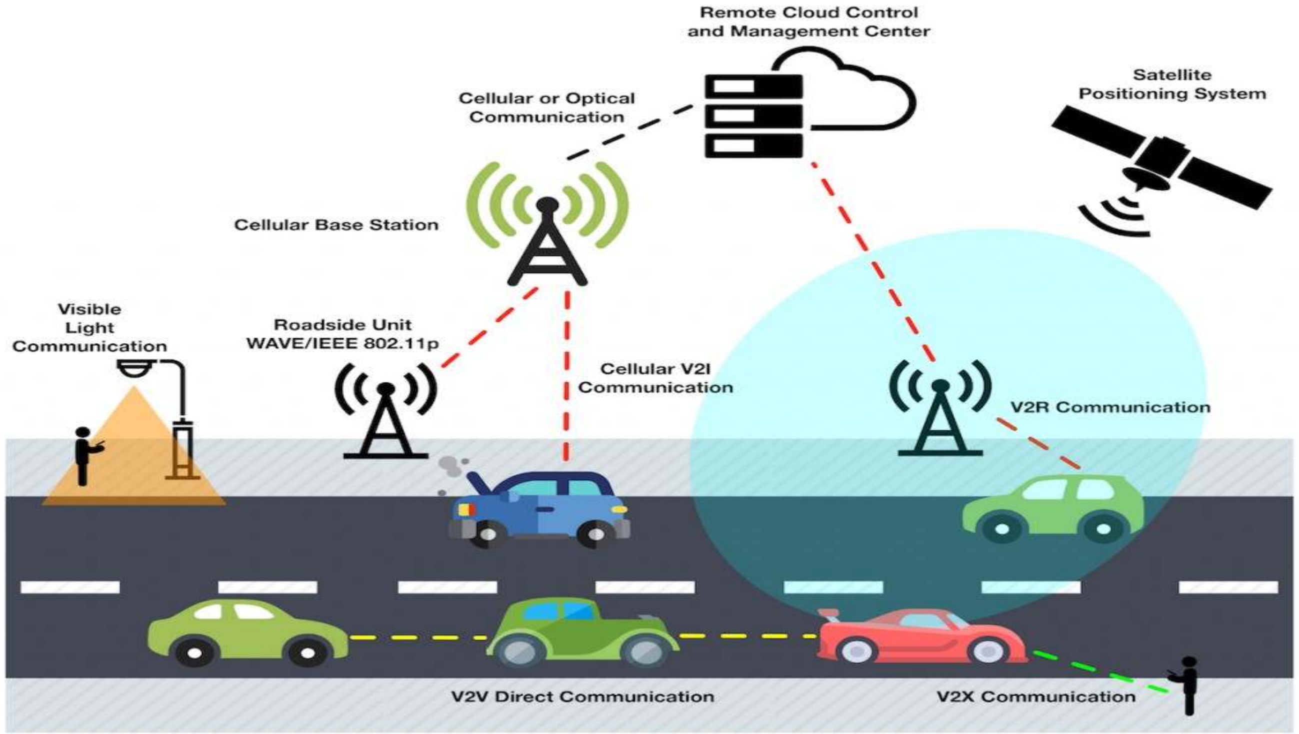
Introduction to the Project

Goal of the Project

- Connected driving with Cooperative Perception using Car2X Technology.

- The project aims to provide the vehicle with the perception of its environment e.g., Map, Accident ahead, lane closure, Obstacle etc.

- Along with that the vehicle can communicate with RSU for CPM/Mapem messages



Introduction to the Project

Goal Breakdown for Car2x Project

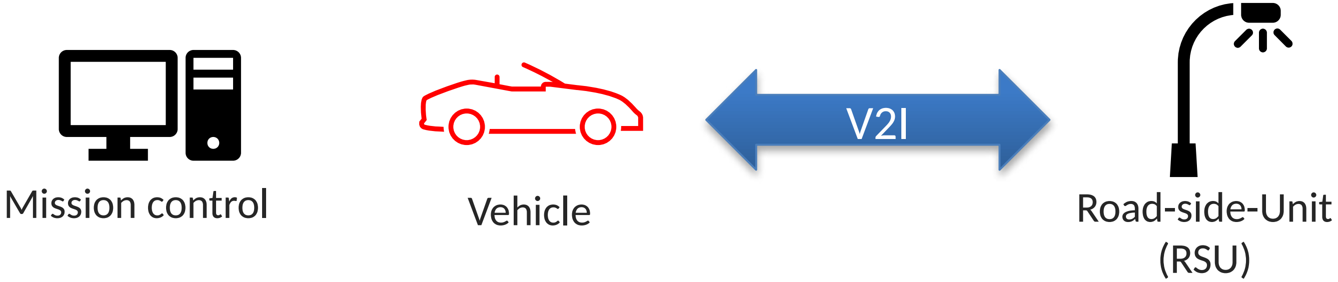
- Working with Perception development kit to create container for the Perceived objects using the Vehicle mounted Radar.

- Realization of CPM Message Communication with On-Board-Unit (OBU) – Cohda box in Vehicle.

- Collection of Map Data for the Road topology around.

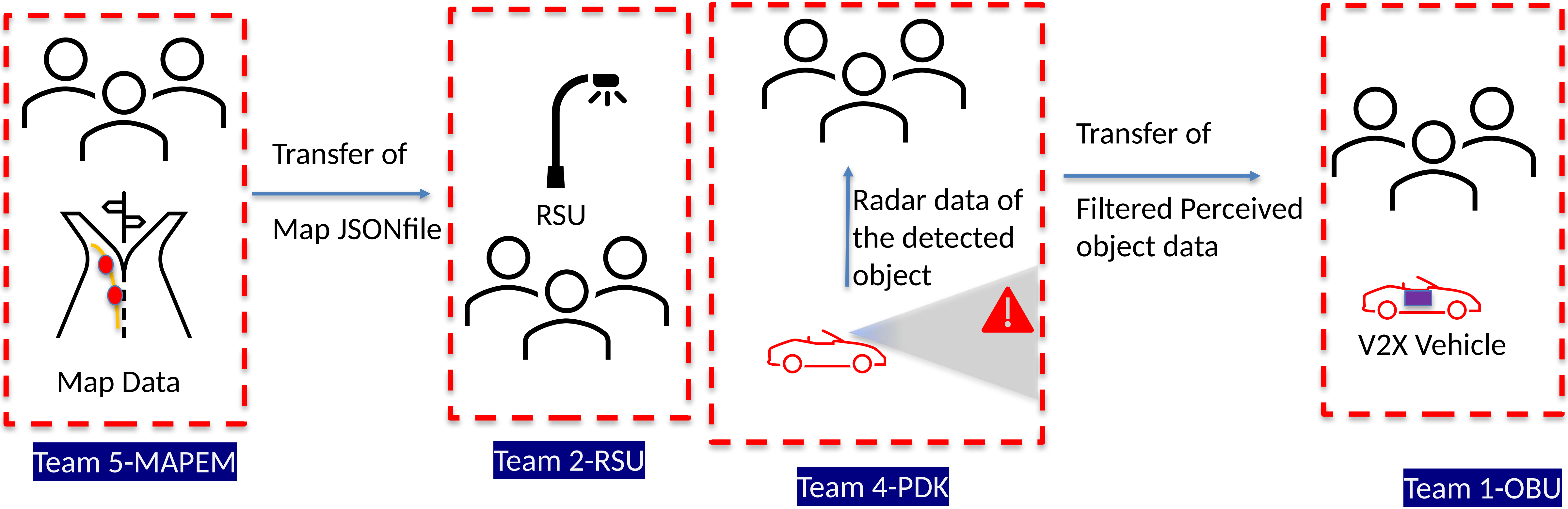
- Development of Mapem data and later Road-side-unit (RSU) can communicate Mapem data with V2x capable Vehicle.

- Development of a Use-case Scenario to realize the working of the overall project



Introduction to the Project

Respective Team Working

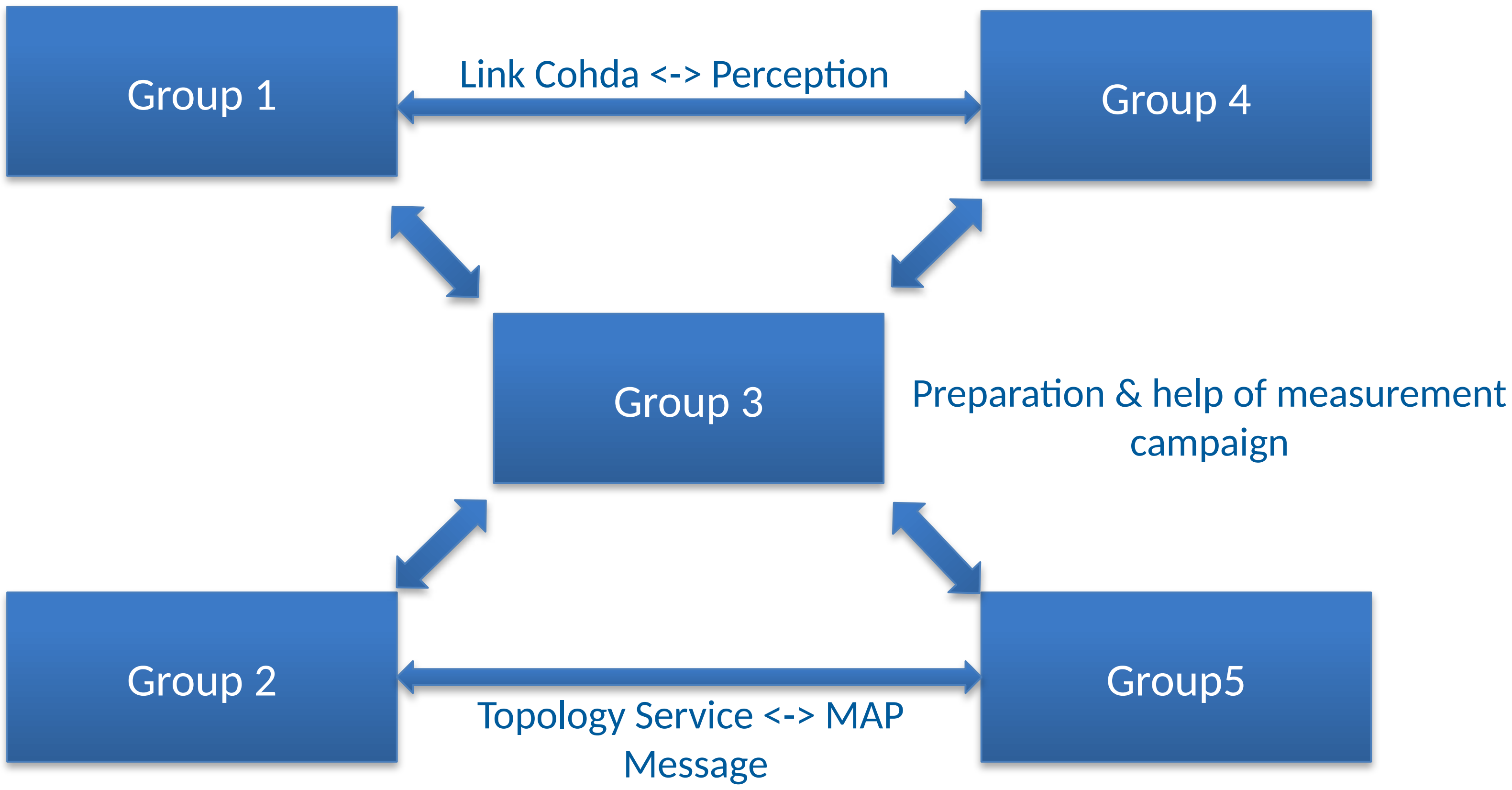


- Team 2 and Team 5 work in collaboration to ensure the MAPEM data being sent from RSU to the Vehicle

- Team 1 works on the OBU (Cohda Box) for the realization of CPM message with Filtered data information from Team 4.

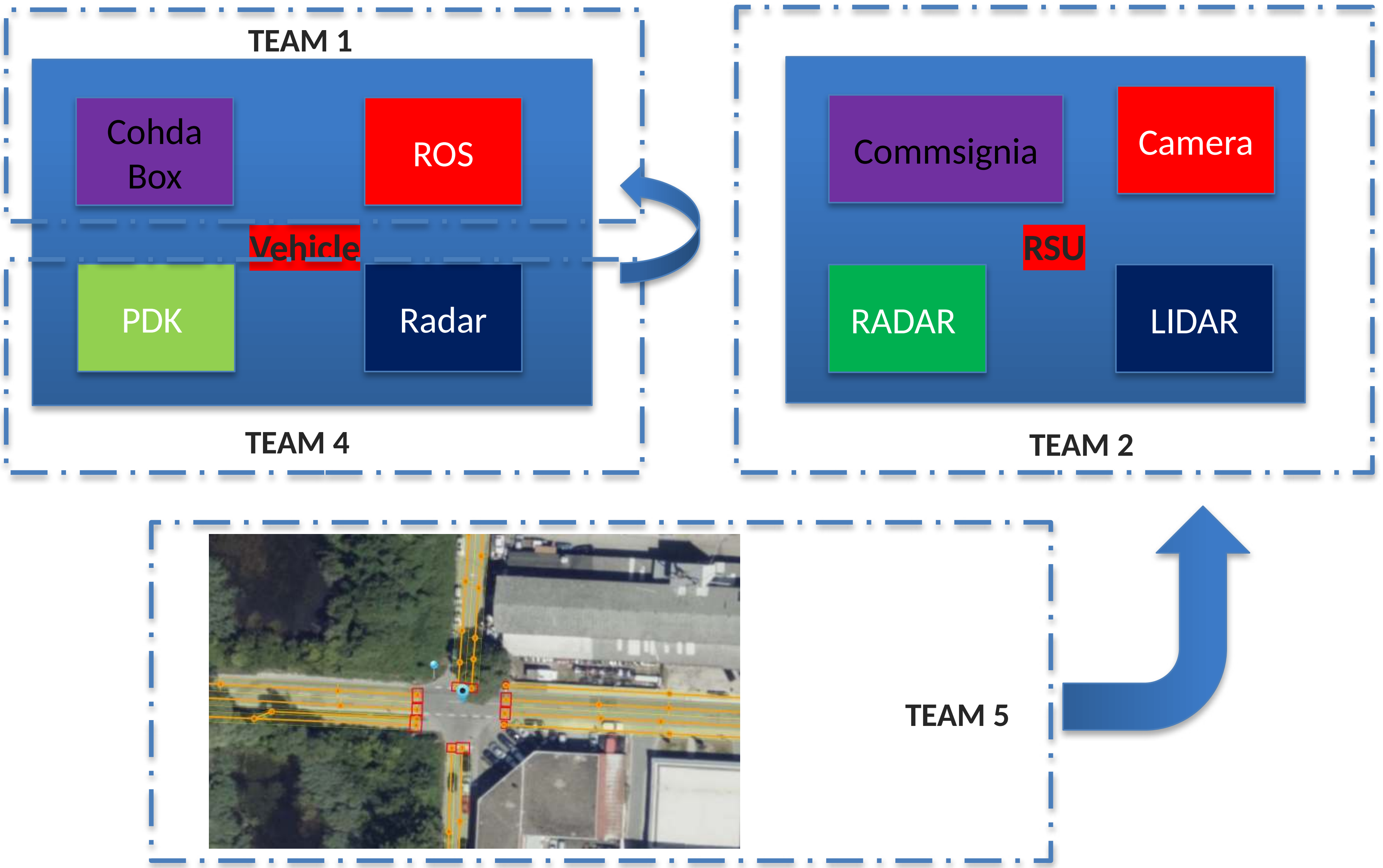
Introduction

Overall Working



Introduction to the Project

Project Architecture



TEAM 4

OBJECTIVES

1

DATA COLLECTION

2

SUBSCRIBING & FILTERING

3

VISUALISATION

4

TRANSFERRING

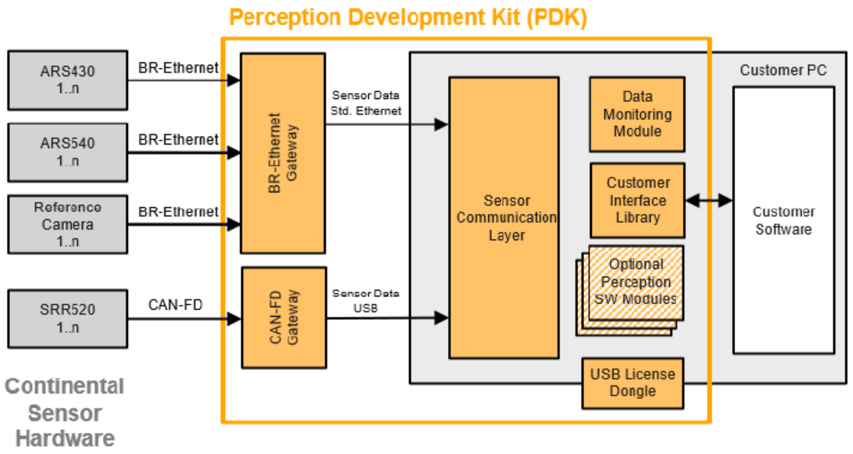
SOFTWARE’s & TOOLS

VM Ware Workstation

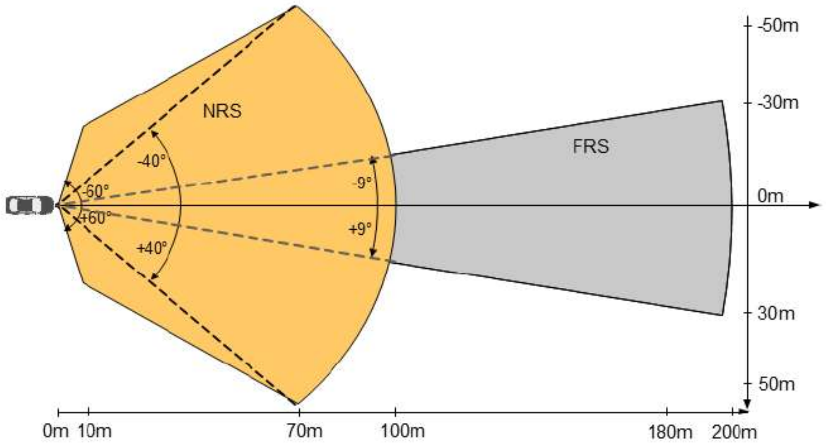
Ubuntu

ROS : Melodic

Perception Development Kit



System Concept Radar + Camera (Exemplary Illustration)



Radar Field of View (Exemplary Illustration)

CHALLENGES AND SOLUTIONS

1. DATA COLLECTION
2. SUBSCRIBING & FILTERING
3. VISUALISATION
4. TRANSFERRING

DATA COLLECTION

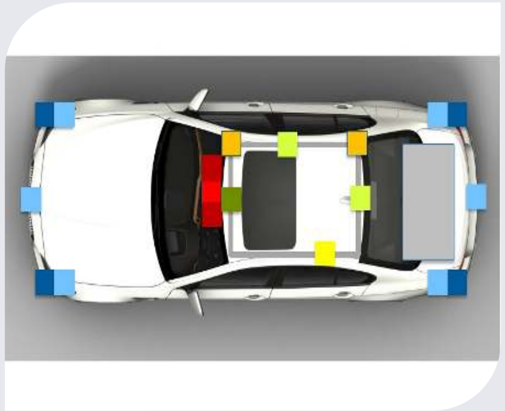
Test Vehicle

DATA COLLECTION

Test Vehicle

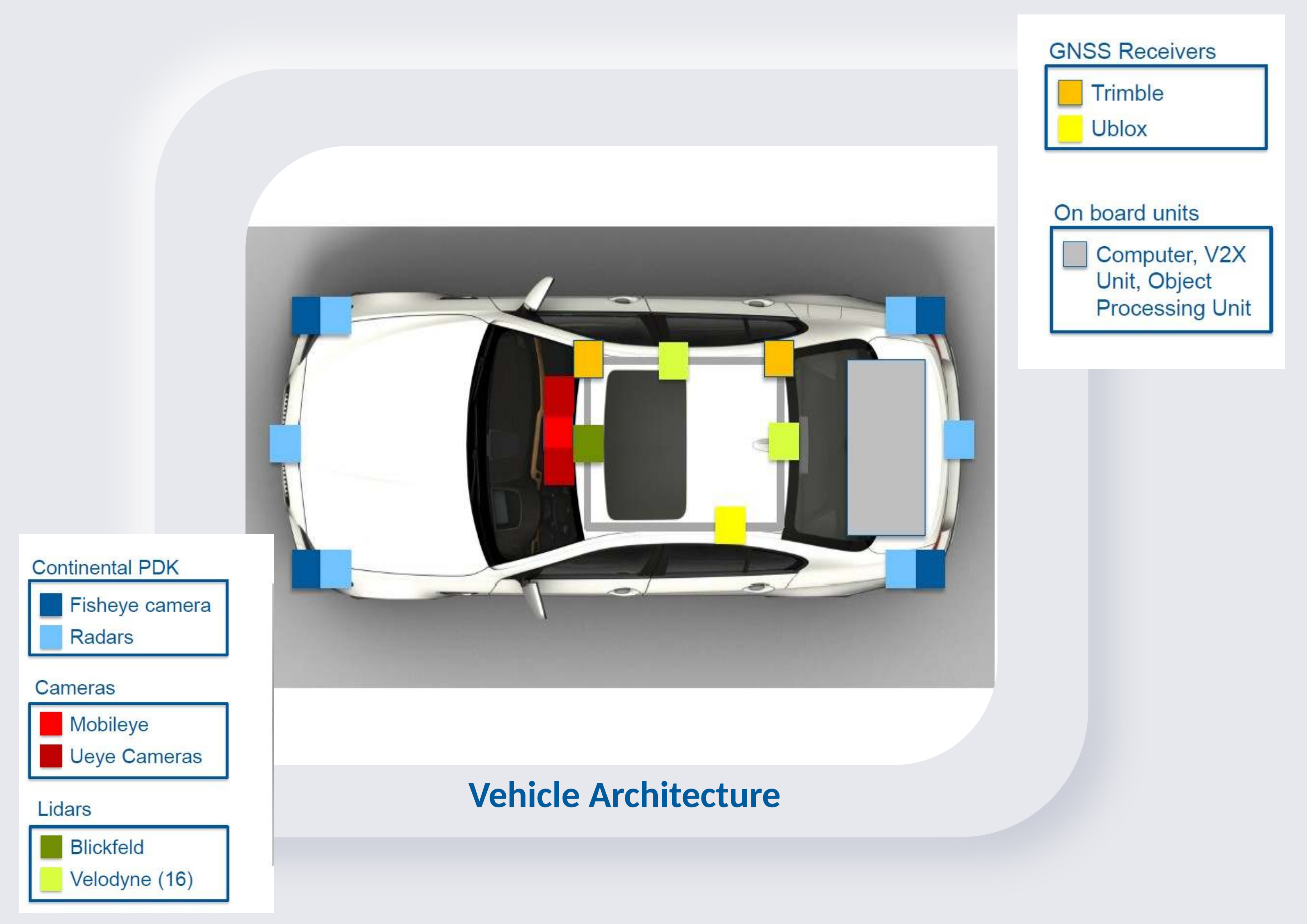
DATA COLLECTION

Test Vehicle



Vehicle Architecture

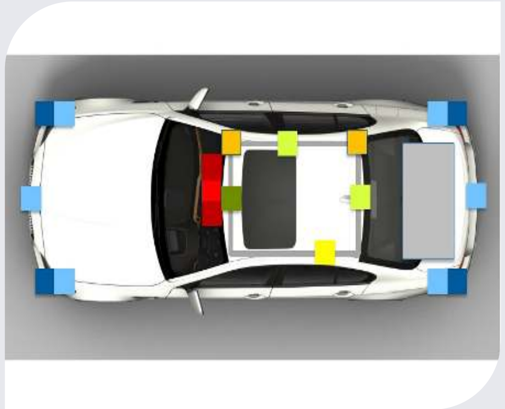
DATA COLLECTION



Vehicle Architecture

DATA COLLECTION

Test Vehicle



Vehicle Architecture

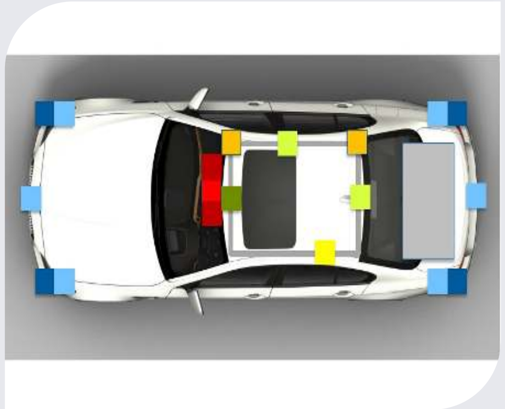
V2X Kit

DATA COLLECTION

V2X Kit

DATA COLLECTION

Test Vehicle



Vehicle Architecture

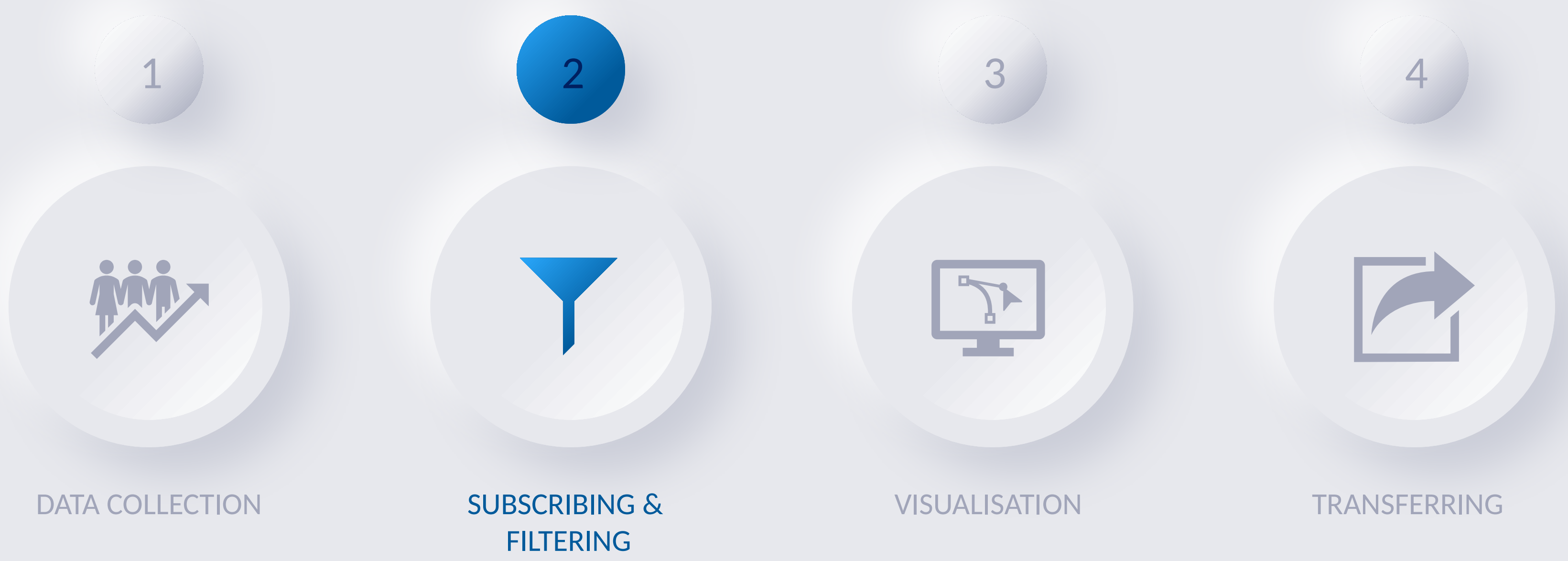
V2X Kit

Radar Data

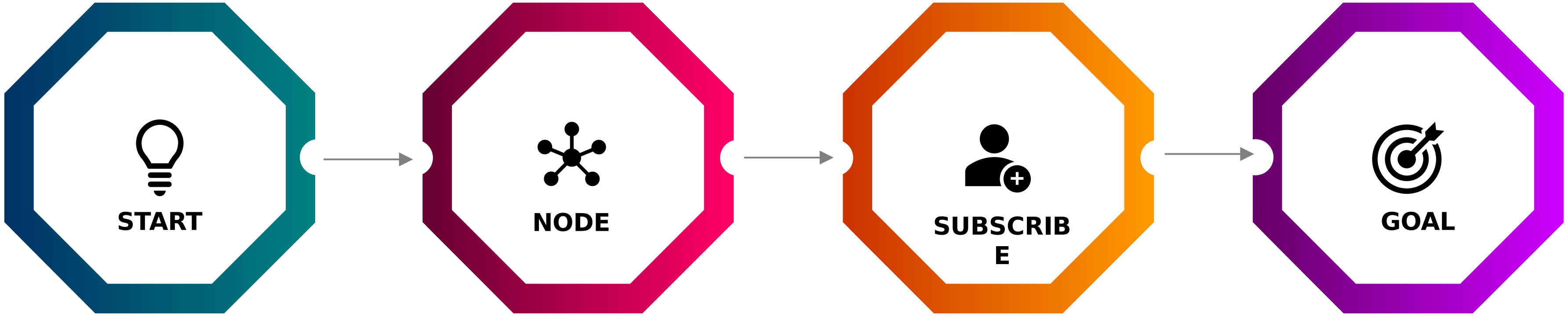
DATA COLLECTION

Radar Data

CHALLENGES AND SOLUTIONS



SUBSCRIBING AND FILTERING



Cloned "pdk_ros_msg" package from github

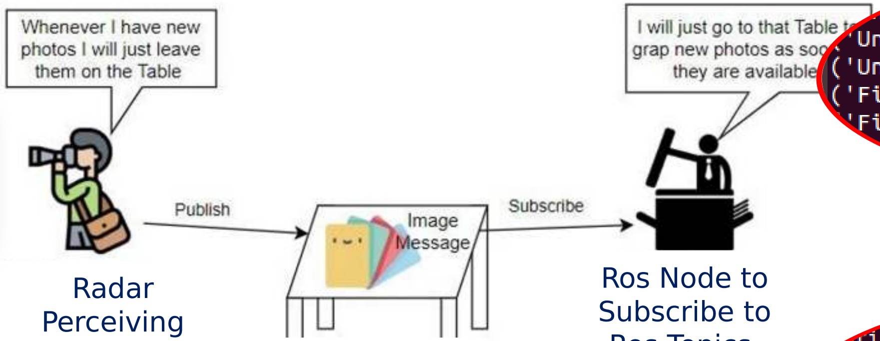
ROS Node created in Python (pdk_tracking.py)

Subscribed to "/pdk/tracking" topic

Filtered the Data based on Distance and Object Score

Set filter Criteria – Object score > 0.9 and distance < 50

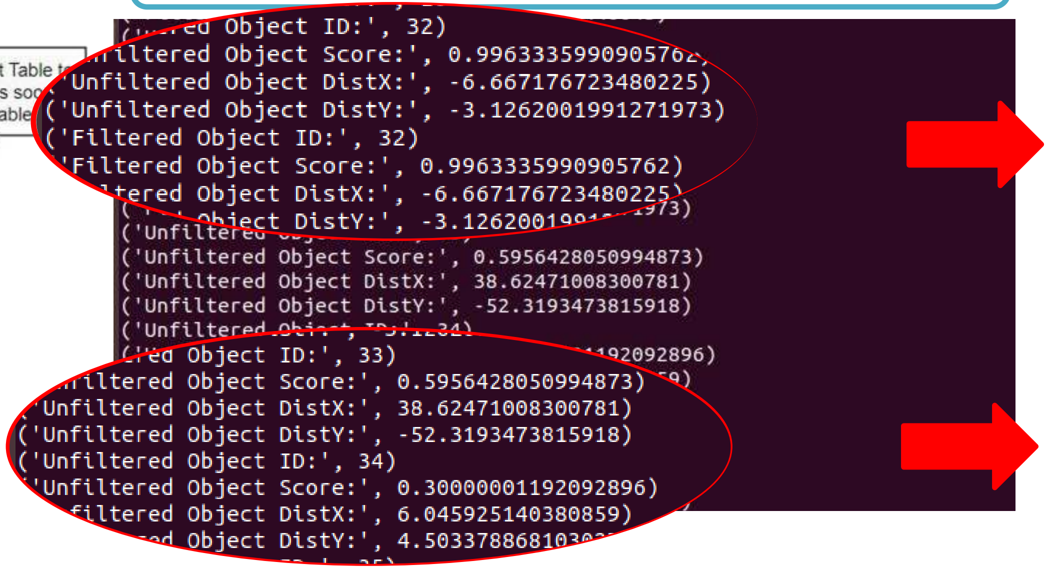
Analogy



Radar Perceiving Objects

PDK Tracking Topic

Ros Node to Subscribe to Ros Topics



Obj ID 32- filtered as well as Unfiltered

Obj ID 33 & 34- Only Unfiltered

CHALLENGES AND SOLUTIONS

1 DATA COLLECTION

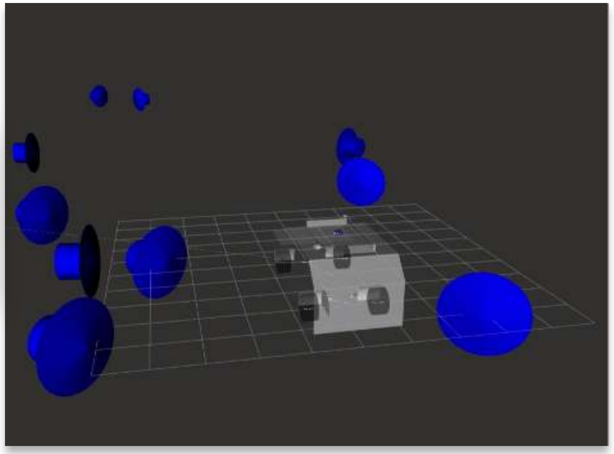
2 SUBSCRIBING & FILTERING

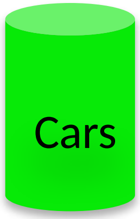
3 VISUALISATION

4 TRANSFERRING

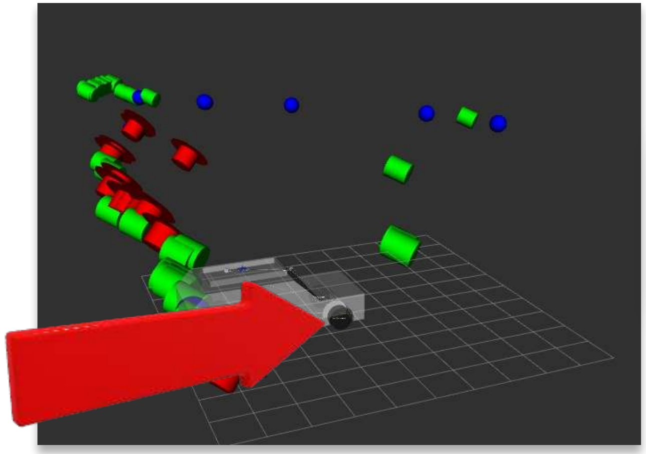
VISUALISATION

Filtered Data

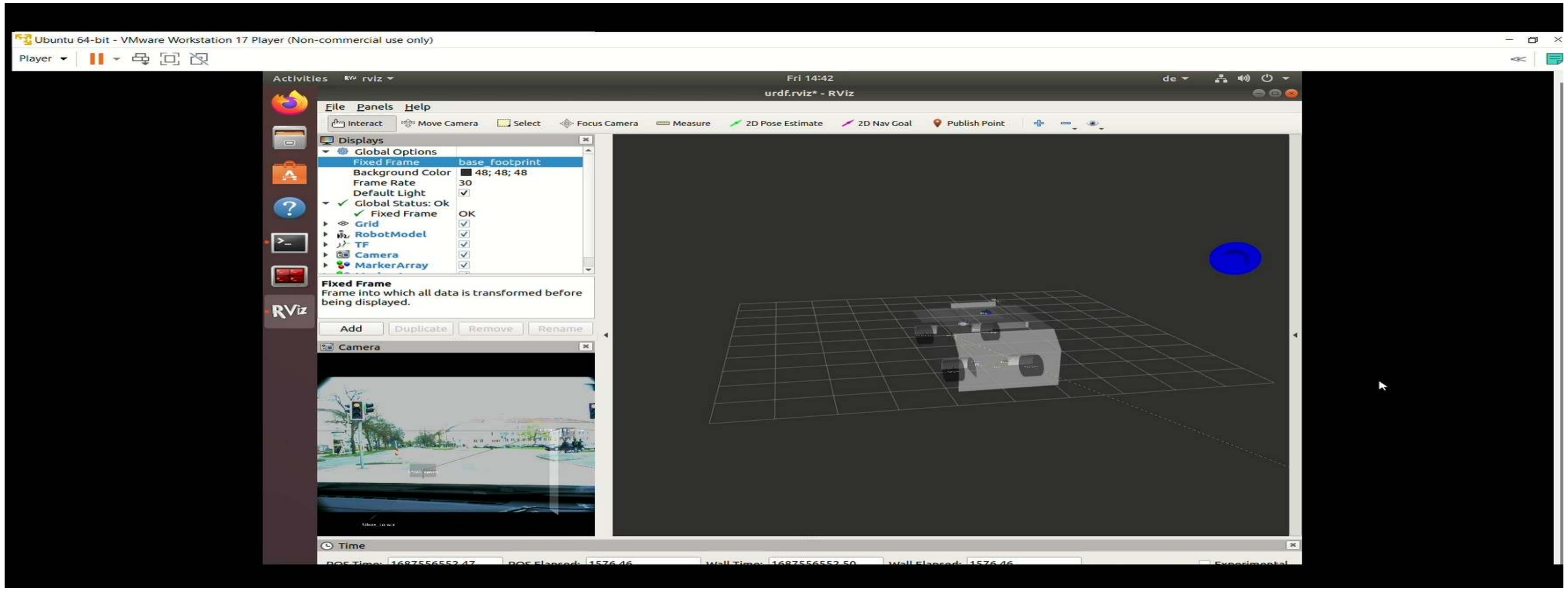




Humans



Object Classification



CHALLENGES AND SOLUTIONS

1 DATA COLLECTION

2 SUBSCRIBING & FILTERING

3 VISUALISATION

4 TRANSFERRING

TRANSFERRING


```
---
header:
  seq: 1061
  stamp:
    secs: 1682064275
    nsecs: 382606167
  frame_id: "base_footprint"
u_ObjId: 8
f_DistX: 0.641784548759
f_DistY: 7.41600322723
f_VrelX: -8.08744621277
f_VrelY: -2.00306034088
f_ArelX: 4.91915512085
f_ArelY: -1.36974155903
f_DistXStd: 0.601692199707
f_DistYStd: 0.673471868038
f_VrelXStd: 0.293487638235
f_VrelYStd: 0.175344899297
f_ArelXStd: 1.38022518158
f_ArelYStd: 0.901765882969
f_LDeltaX_left: -0.700744748116
f_LDeltaX_mid: -0.367649227381
f_LDeltaX_right: 0.799088120461
f_LDeltaY_left: 0.646389007568
f_LDeltaY_mid: -0.836938381195
f_LDeltaY_right: -0.574936389923
f_RCS: -4.78357124329
f_ObjectScore: 0.945010304451
u_LifeCycles: 12
f_VabsX: -1.03724229336
f_VabsY: -2.01754593849
f_AabsX: 4.91915512085
f_AabsY: -1.36974155903
f_VabsXStd: 0.293458402157
f_VabsYStd: 0.175302073359
f_AabsXStd: 1.38022518158
f_AabsYStd: 0.901765882969
---
```

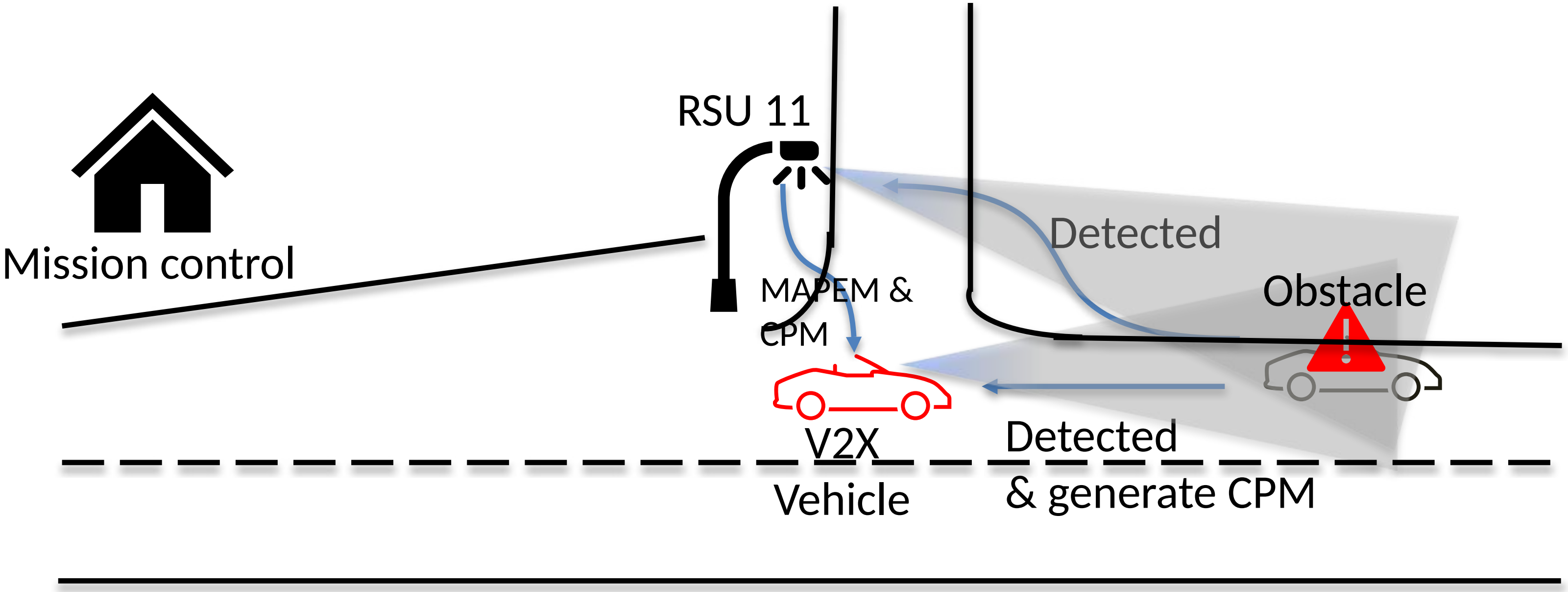

PDK Tracking Output


```
numberOfPerceivedObjects: 1
perceivedObjects:
  -
    objectID: 1
    measurementDeltaTime: 1682064289
    position:
      x_cord:
        value: -805
        confidence: 0
      y_cord:
        value: 83
        confidence: 0
      has_z_cord: False
      cartesianVelocity:
        xVelocity:
          vel_comp_value: -1638
          speed_confidence: 0
        yVelocity:
          vel_comp_value: -414
          speed_confidence: 0
    cartesianAcceleration:
      xAcceleration:
        value: 1
        confidence: 0
      yAcceleration:
        value: -6
        confidence: 0
      has_zAcceleration: False
  objectAge: 157
  objectPerceptionQuality: 0.950932443142
```


Team 1

Scenario

Team 3 Working



- In our case we have one object which is Bicycle.
- Then RSU is continuously sending MAPEM message.
- The Trailer(Obstacle) is stopped near by RSU 11 and is detected by RSU and CPM message is sent by the RSU
- Obstacle Detected by the vehicle PDK unit (Team 4 Working) and Generates a CPM Message

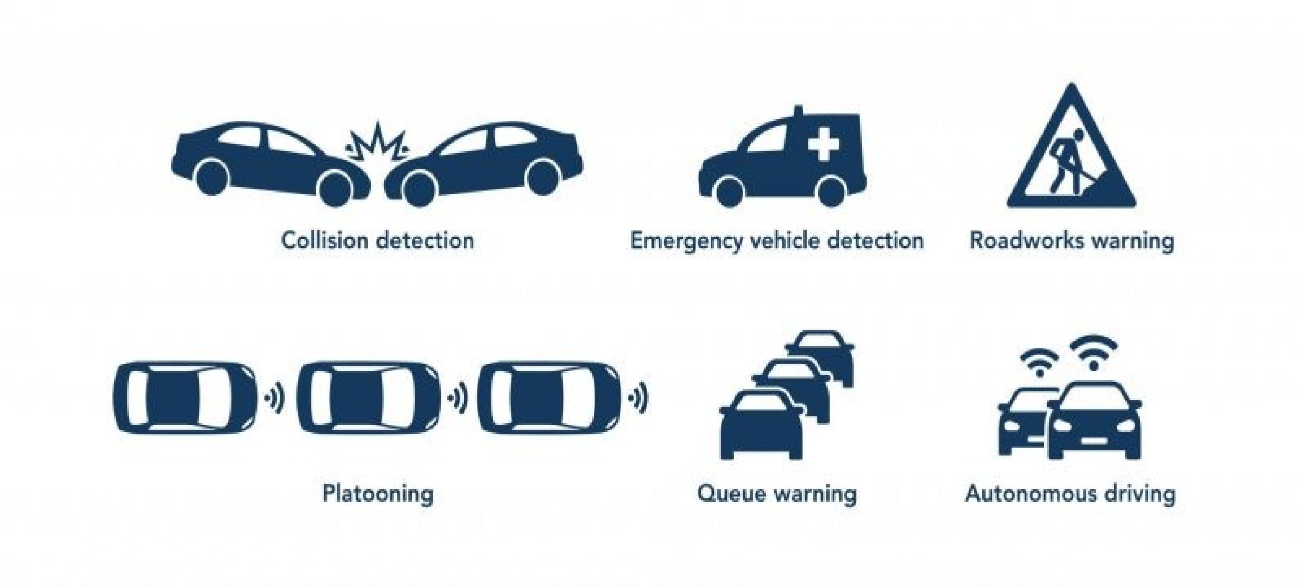
Scenario

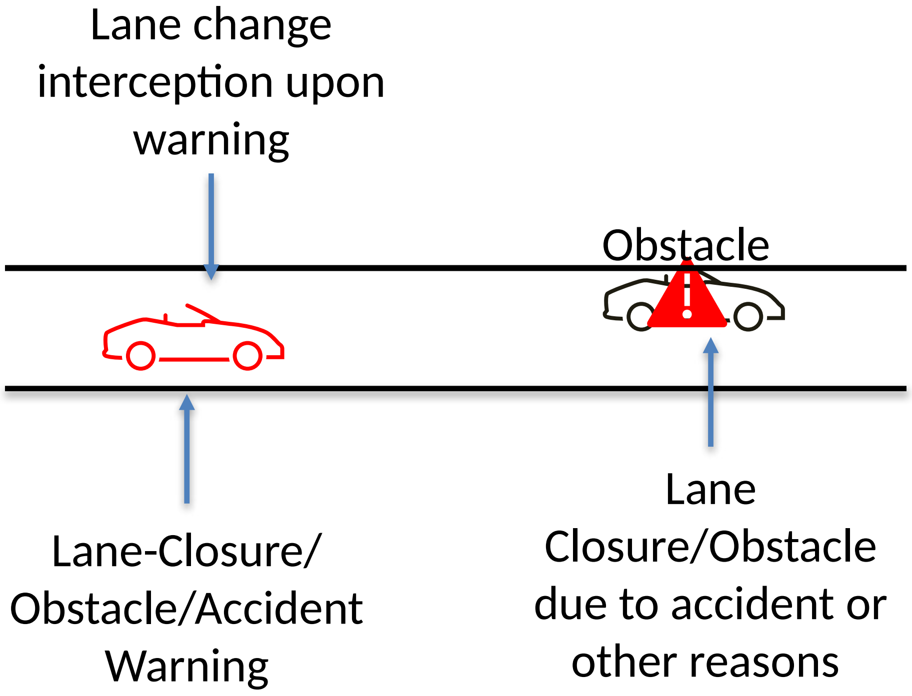
Trailer Location

- Red block indicates the position of the Trailer(Obstacle) for the visualizing the Use-case scenario.
- It is 5m from the intersection and placed tightly closed to the footpath to avoid the congestion.

Usecase related to the scenario

- Obstacle Warning and Avoidance
- Lane Closure warning
- Accident Warning
- Lane Change warning





Learning Outcomes

- MAPEM message and the indispensable requirement of such detailed road topology.
- Vehicular communication Standards: ETSI 103 301, SAE J2735, and ISO 19091
- C++ Parsing, Geonetworking Protocols and ROS environment
- Environmental Perception, Visualisation with Sensor data sharing and processing